

ScrAlter: An LLM-powered Authoring Tool to Support Cultural Heritage Experts in Interactive Narrative Game Scriptwriting

Background & Motivation

Interactive narrative games have been proven to be an attractive and effective way for visitors to experience cultural heritage sites while learning. Designing the game scripts is a critical step that requires domain expertise in history and culture. However, the scriptwriting process lacks technical support. To this end, we implemented ScrAlter, an LLM-powered authoring tool for game scripts, utilizing RAG to help understand material and create content.

Objectives

Following the Research through Design (RtD) paradigm:

- Understand the need for experts when creating narrations for CH sites
- Assess the usability and UX of ScrAlter
- Evaluate workflow support

Research Questions

- What kinds of technical support do cultural heritage experts need when designing interactive narrative game scripts?
- How does ScrAlter support cultural heritage experts in their authoring workflows?
- Are game scripts created with ScrAlter usable for deployment (comparable to human-created ones)?

Design Overview

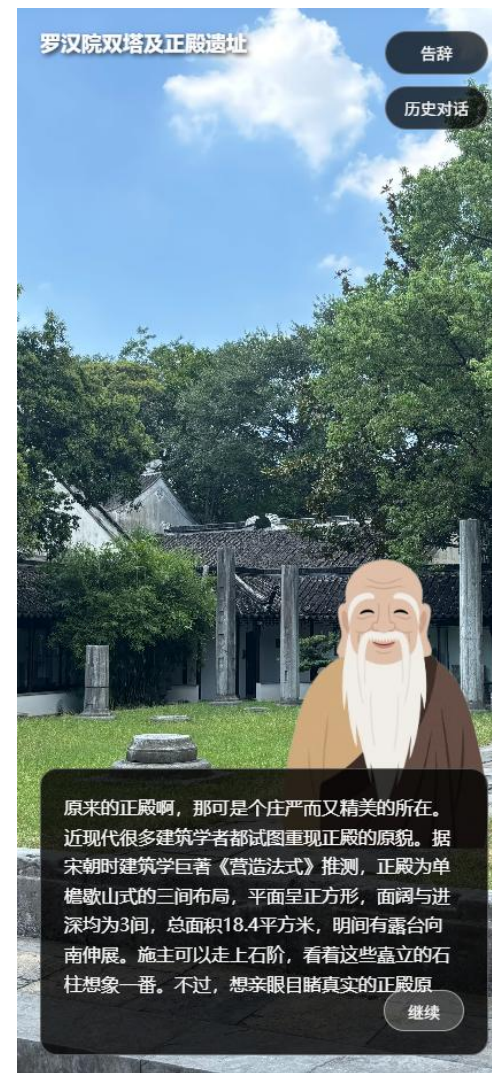
- **User profile:** Cultural heritage experts with related knowledge
- **Scenario:** Interactive narrative game script creation for cultural heritage sites.
 - ◆ Act as a tour guide, providing information on different themes and at various levels of detail.
 - ◆ Players will have conversations with NPCs, connecting with sites and ages.
 - ◆ Players can choose their interested options (questions) and get more detailed answers.
 - ◆ Example: <https://interactive-narrative-game-ch.vercel.app/>



NPC



Player options

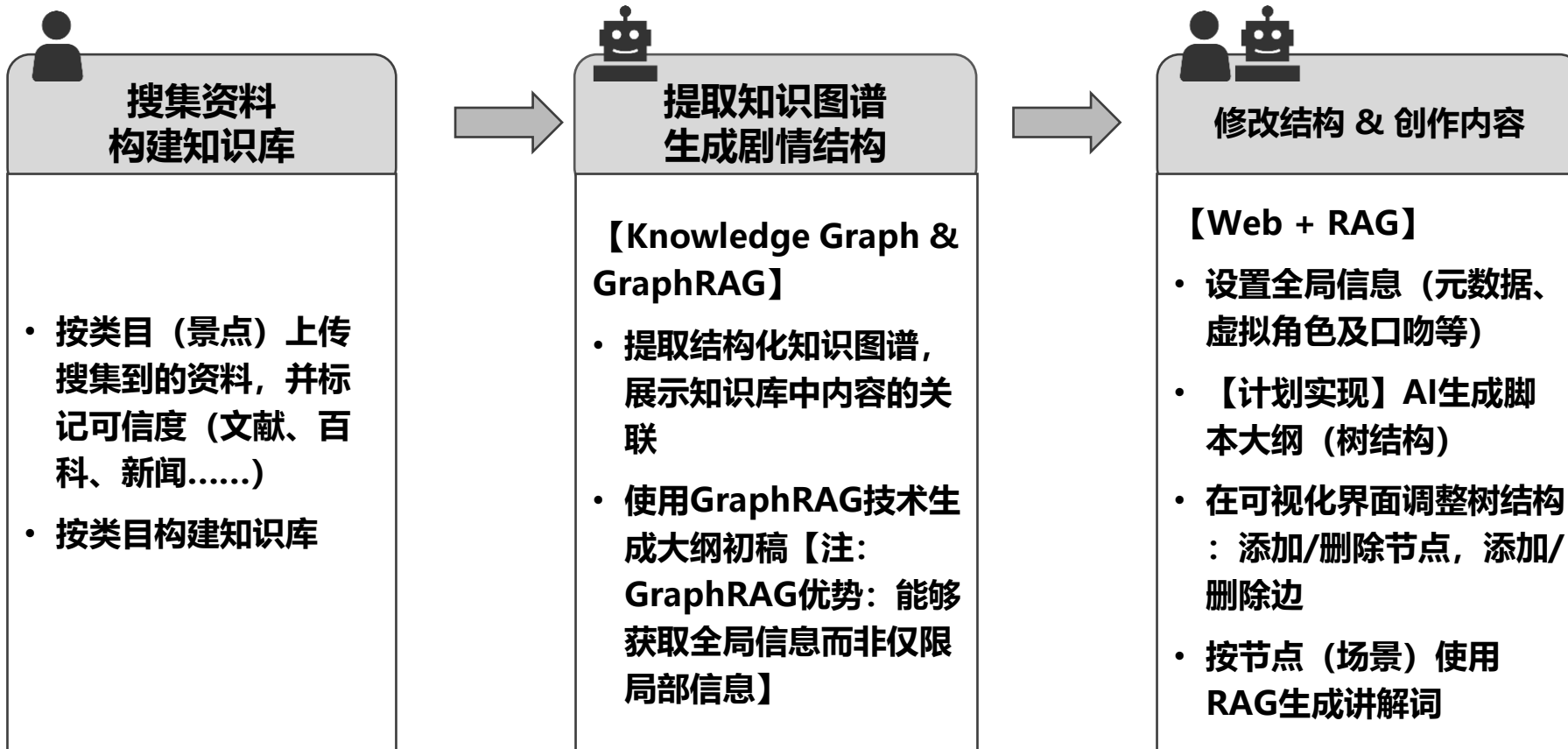


Background switching

Design Overview

- **Problems:**
 - ◆ Script design requires extensive searching and reading materials
 - ◆ Manually creating interactive branches is laborious (not so user-friendly)
 - ◆ Cultural heritage experts usually lack technical support
- **Solutions:**
 - ◆ Content level: Build knowledge bases with related materials; Create introduction text via RAG given a player option (query) and prompts from creator.
 - ◆ Structure level: Visualize tree structure; Allow direct editing on the graph.

Current Functions





知识库管理

+ 新建知识库

知识库数: 1

知识库名称	类目	文件数	切片数	操作
📄 twin_pagoda	双塔	10	119	🔗 图谱 🗑️

提取结构化信息

文件管理

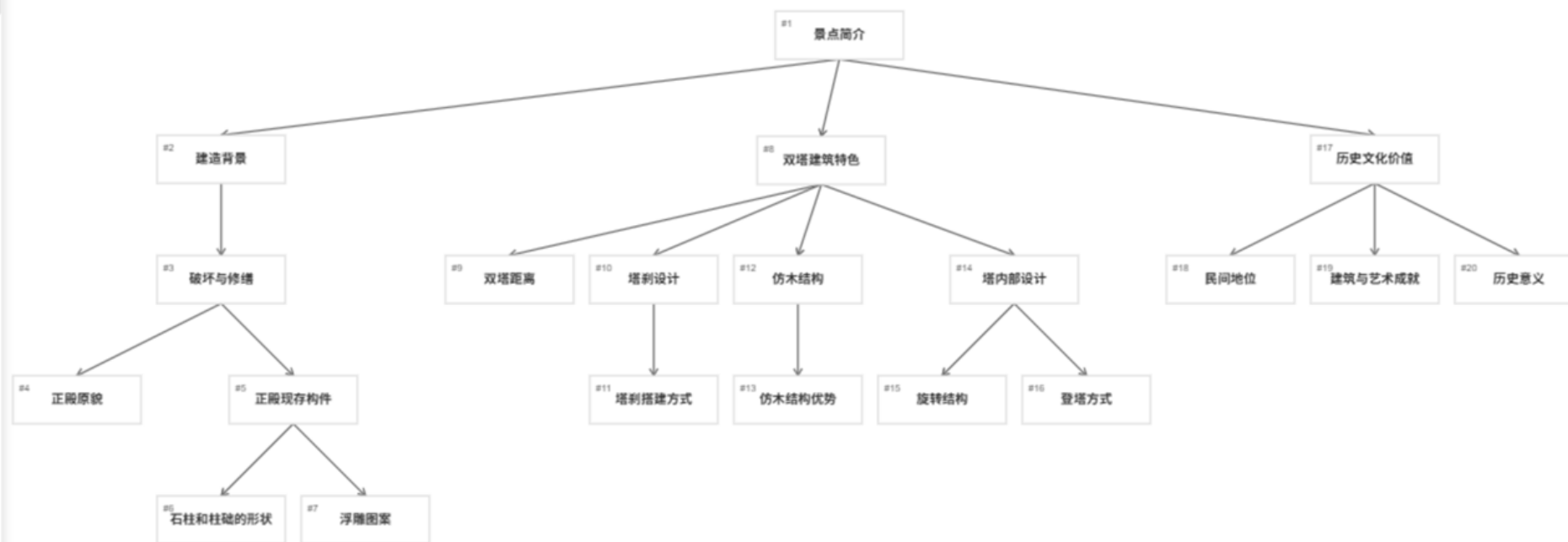
📁 上传文件

类目数: 1 总文件数: 10 未标记: 0

▼ 📁 双塔 10 个文件

标记了资料来源的类型
代表可信度

文件名	类型	可信度	操作
📄 千年双塔亲密对望.docx	非结构化	网络文章	▼ 🗑️
📄 双塔禅寺.docx	非结构化	网络文章	▼ 🗑️
📄 宋代童趣图浮雕艺术研究_吴晓楠.pdf	非结构化	文献资料	▼ 🗑️
📄 梁思成破译.docx	非结构化	网络文章	▼ 🗑️
📄 罗汉院双塔及正殿遗址.docx	非结构化	百科知识	▼ 🗑️
📄 苏州双塔.docx	非结构化	百科知识	▼ 🗑️
📄 苏州双塔寺.docx	非结构化	网络文章	▼ 🗑️





节点 #1: 景点简介

节点编辑

关键词

景点简介

用户选项

这里是什么地方？能给我介绍一下吗？

摘要

罗汉院双塔与正殿遗址的历史地位、建造背景与整体价值概述

内容

施主，你问得好。这里便是罗汉院双塔及正殿遗址，苏州城内一处穿越千年的圣地。双塔始建于北宋太平兴国七年，历经沧桑，至今仍傲然挺立。它们不仅是建筑学上的奇迹，更是见证了无数文人墨客的风雅往事。东塔与西塔相距不过二十米，却能在如此近的距离下稳固屹立千年之久，足见古人的智慧与技艺之高超。

而正殿虽已不复昔日辉煌，但从残存的石柱础上仍可窥见其当年的恢弘气势。每当晨曦初露或夕阳西下时，漫步其间，仿佛能听到历史的低语，感受到那份跨越时空的文化积淀。这不仅仅是一座寺庙的故事，更是整个江南文化脉络中不可或缺的一环啊。

生成内容

保存

创作者的要求

RAG生成内容

参考资料

便于查看参考资料

来源：苏州双塔寺.docx 网络文章

相关度：33.2%

双塔寺的西边是定慧寺，东边则是清代设立的贡院，这里是贡生（童生）考秀才的地方。贡院占地约六亩，四周有围墙，正门两侧用木栅栏，形似衙署，门口还放着一对石狮，显得格外威严。院内有广场，供停轿系马，广场两端竖立着旗杆，并建有鼓乐亭，上下两层，上层放炮，头炮准备，二炮进考场；下层吹打。每逢考期，这里张灯结彩，鼓乐喧天，十分热闹。两边设有考棚，每个考生一间，仅容一桌一椅。清末科举停办后，贡院就此冷落，成为荒场，后来建成了民居。

民间传说中，苏州城内的文房四宝齐全，府城是笔墨盘，双塔是两支笔，钟楼（俗名“方塔”、“文星阁”，在苏州大学内）是墨，孔夫子巷内原有水池，为墨池，苏东坡曾在此洗砚。有这等巨大的笔墨盘写文章，苏州文人在全国考试中屡屡夺魁，仅有清一代，苏州府就出状元二十六名，其中一人还连中三元。